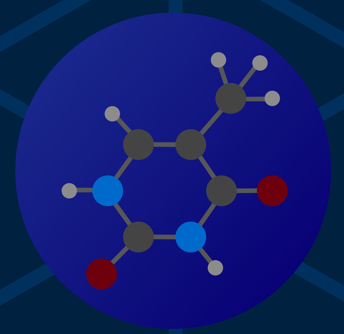
WHAT IS **3D PRINTING?**

3D Printing is a collective term, used to identify the Additive Manufacturing process.
In this process, a 3D object is built by successive layering of single or multiple materials.

WHAT ARE THE **APPLICATIONS?**

Design and Prototype

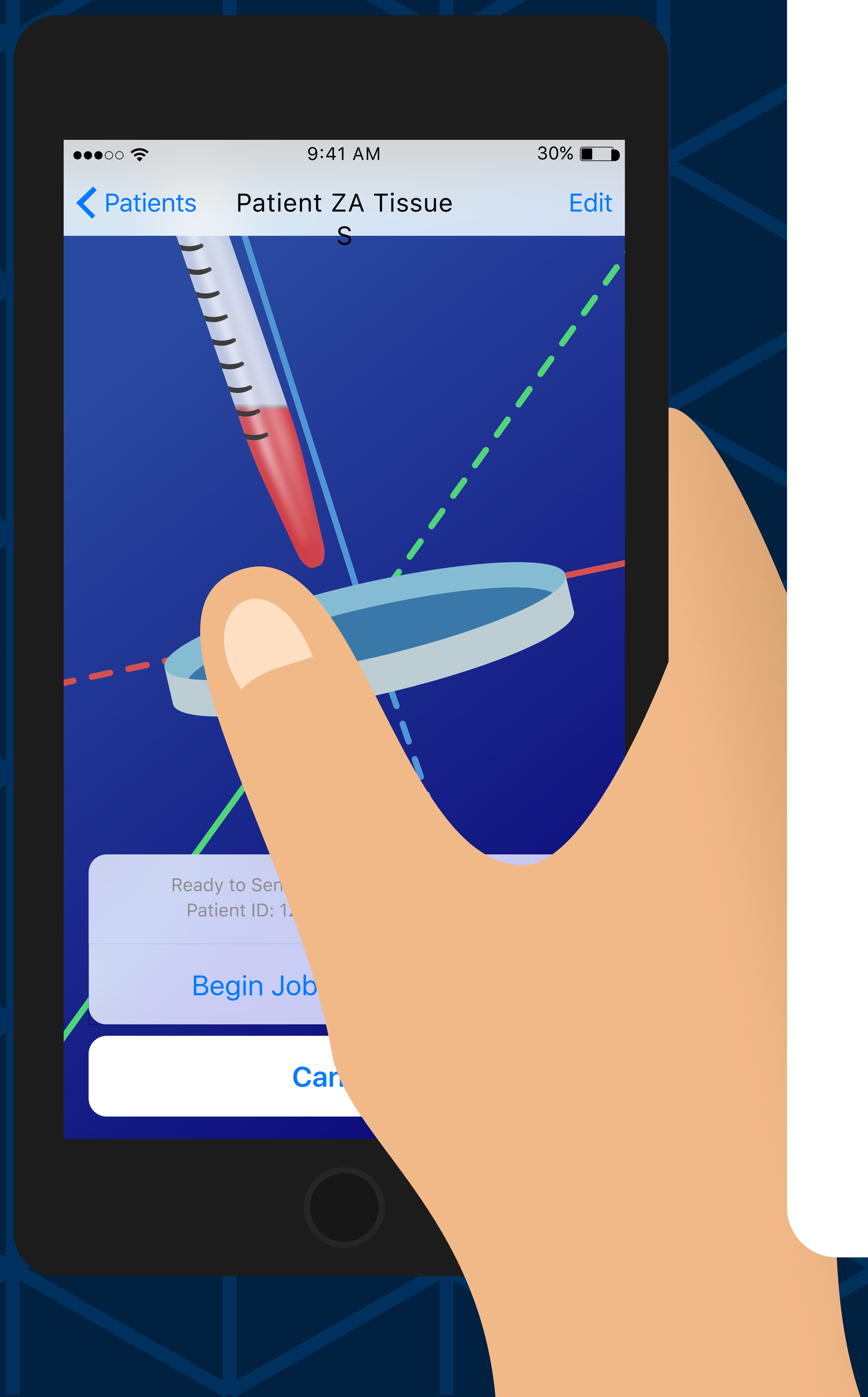
Visualization



Education and Learning

Medical Applications

Customized Prosthetics, Corrective equipments, and Bioprinting of Donor Organs are some trivial areas where 3D Printing may be used.
Research is ongoing to make it possible to use chemical inks to print medicine.
Similarly, 3D printing has been considered as a method of implanting stem cells capable of generating new tissues and organs in living humans.



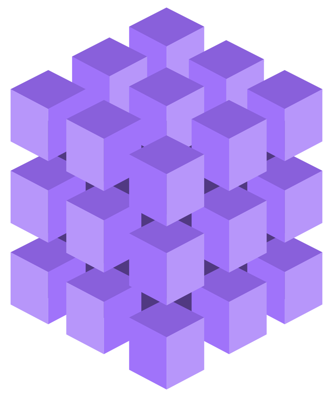
Cluster Innovation Centre
University of Delhi

About Cluster Innovation Centre

The Centre has been designed to seek and drive innovations from industrial clusters, village clusters, slum clusters and educational clusters.

We strive to stream relevant ideas and programmes stemming from the above mandate into its learning and research programmes.

The CIC has pioneered the concept of a Meta College as well as a Meta University and runs highly innovative state of the art learning and research programmes.



http://ducic.ac.in

University Stadium
G.C. Narang Marg
University of Delhi
Delhi - 110007

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Everything About 3D Printing

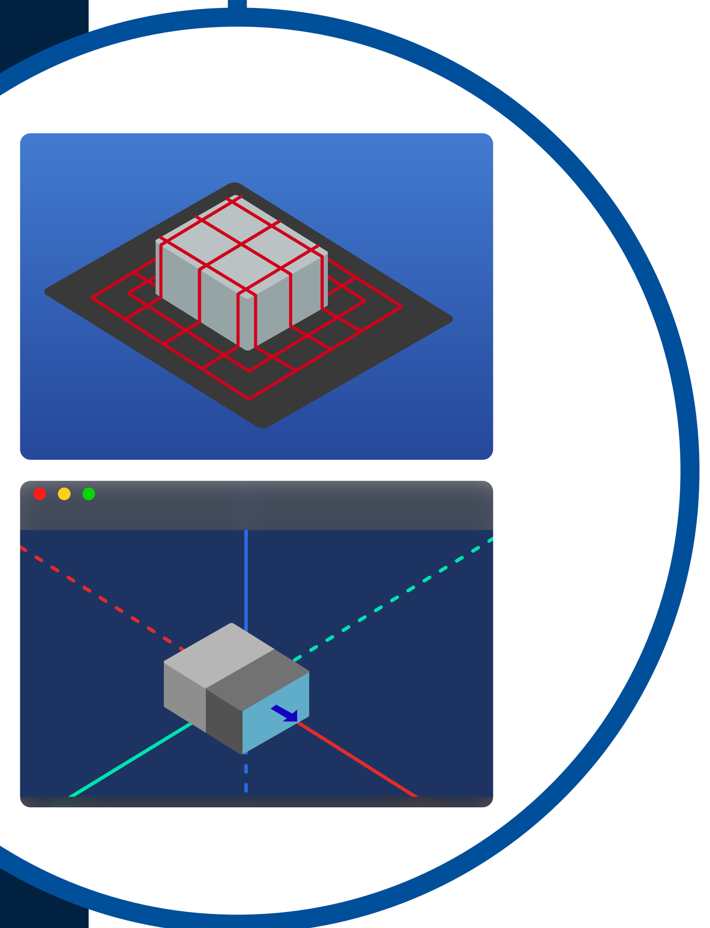
HOW DOES **3D PRINTING** WORK?

1 2 3

Not all 3D Printers are alike. Each one of them are built with specific application in mind. However, all of them have the following three basic concepts implemented.

1 Creating the STL File

Techniques like Laser Scanning, CAD Modelling, CT Scan, and MRI Scanning give us the 3D Spatial information about any object that we wish to either print or display on the computer screen. However, it is infeasible to design a 3D printer that can process such a wide variety of 3D data. Hence, the 3D Spatial Data is converted into the STL (STereoLithography) format. STL is a widely accepted format for 3D Models and is used in all kinds of Computer Aided Manufacturing processes.

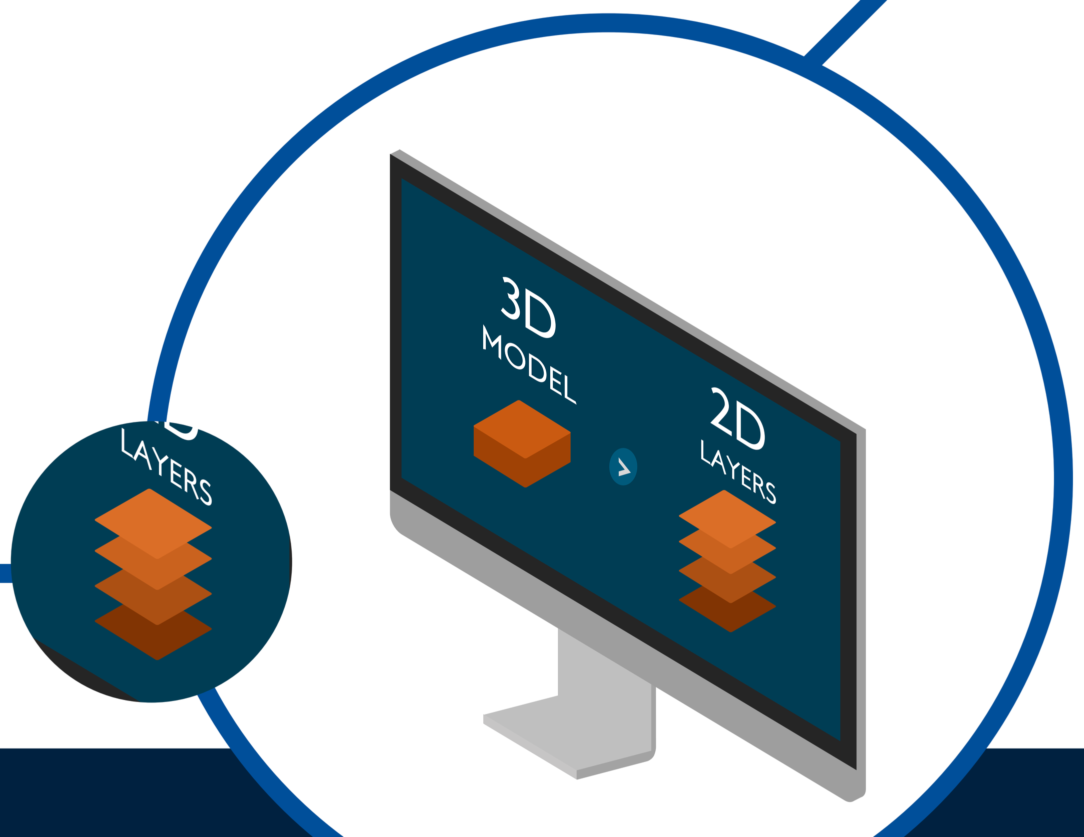


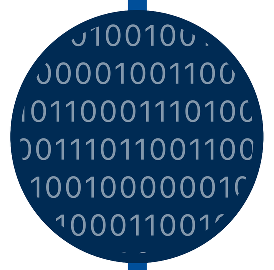
2 Processing the Print Job

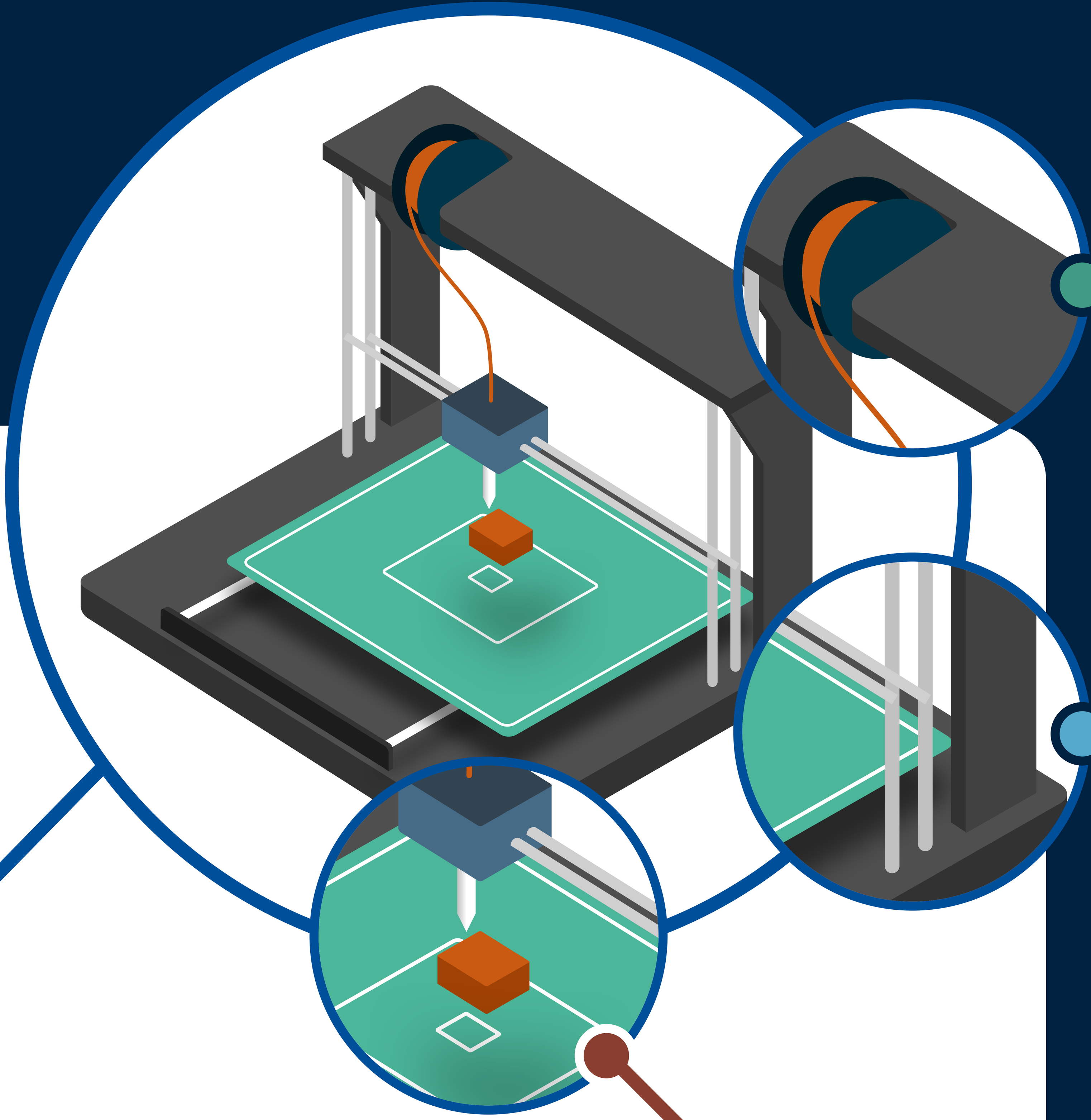
The STL file is then loaded into a printer specific software that slices the 3D Model in STL format to multiple 2D layers.

The thickness of the layer depends on the resolution of the 3D Printer.

The software may also add some removable support structures for overly complicated or unstable 3D Models.



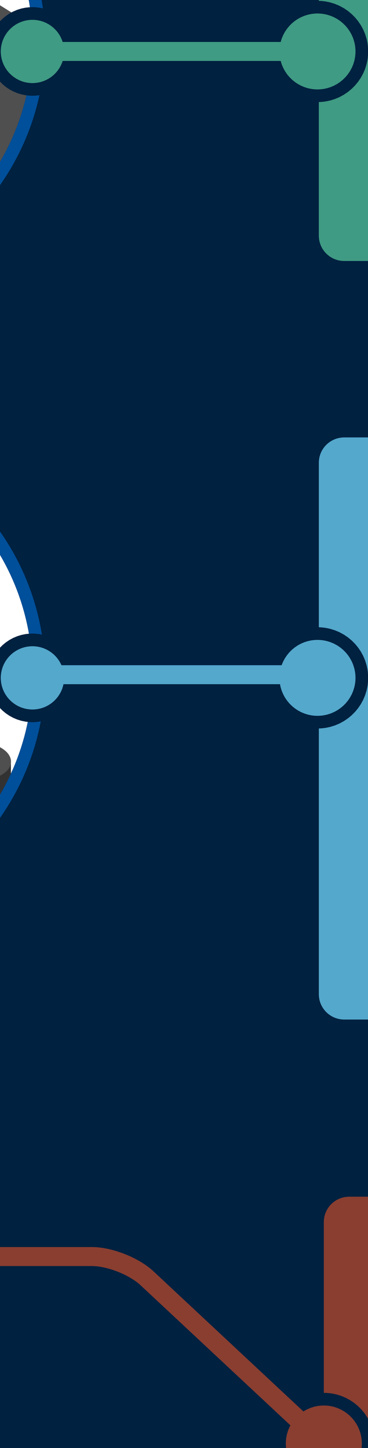




Instruction sets are sent from the Computer to move the print head in any desired direction.

3 Printing Process

The process of printing is very straightforward.
Beginning with the bottom most layer, the printer starts laying down the material.
Once done, the printer lays the next layer over the previous layer which is bonded due to the adhesion property of the Material.
This process is repeated until all the layers have been deposited.



ABS, being impact resistant, tough, and heat resistant, is one of the ideal materials that may be used for 3D Printing.

The 3D Spatial movement of the printing head is achieved by independently controlling each of the three axes. The axes are hooked with a stepper motor that allows highly precise control in each axes.

Close to 200°C, the printing head extrudes the melted plastic and deposits it to form the layer as per the instructions received from the computer.

The objects can be made out of

Plastic
Game Controllers, Robots, Replacement Parts etc.

Metal
Complicated Jewellery, Rings, Keys, etc.

Medical Usage
Bone fixtures, Titanium fixtures, Prosthetics, etc.